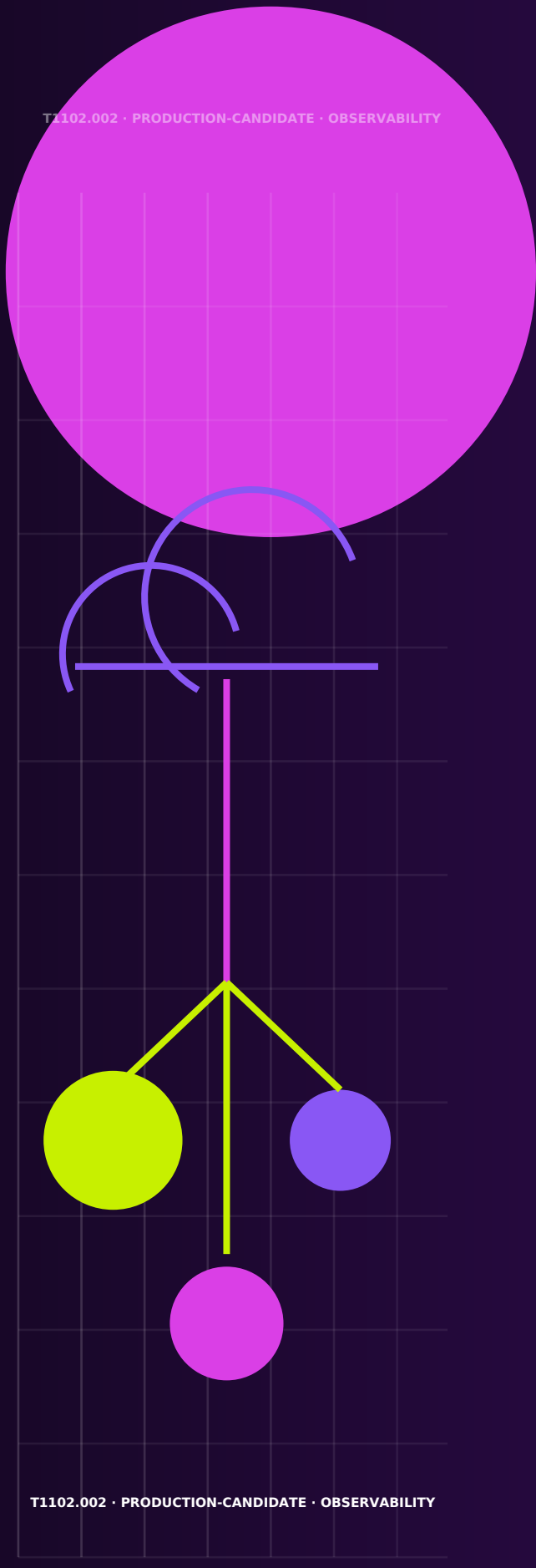
DETECTION THREAT LANDSCAPE

HOLLOWGRAPH - HOLLOWGRAPH CALENDAR C2

access associated with a covert bidirectional channel.



PRODUCTION CANDIDATE

CONFIDENCE: HIGH

REPORT TYPE
campaign

STATUS
Production candidate

PRIMARY PLATFORM
Microsoft Defender XDR

OPERATIONAL VALUE
9 / 10

01

Executive Assessment

Decision-grade summary with explicit evidence boundaries.

ASSESSMENT

DECISION

Retain this package as a production candidate and migrate it to the canonical visual and evidence contract without changing query conditions. Treat every result as an investigative lead until local schema, field population, benign overlap and the legacy claim provenance are reviewed.

LEGACY EVIDENCE RECORD

- The retained legacy source ledger contains 5 normalized references; claim-to-source mapping remains...

ANALYST ASSESSMENT

- The existing logic is retained at production-candidate maturity; repository migration is not...
- The folder and query header map the package to T1102.002 / T1071.001; operational relevance...

DETECTION ASSUMPTIONS

- The named telemetry is collected across the intended asset or identity scope with adequate...
- Projected fields preserve their expected semantics in the target connector and parser. [C002]
- Approved activity, owners and change records are available to analysts during disposition. [C003]

KEY FINDINGS

FINDING 1

The original query conditions and source ledger were preserved during layout migration; legacy PDF SHA-256 fbe37c52686bdbea... is recorded for traceability. [C001]

FINDING 2

The principal observable is Evaluate MicrosoftGraphActivityLogs for hollowgraph calendar c2 using the existing source-derived conditions while preserving projected investigation context. [C002]

FINDING 3

The evidence boundary remains explicit: a match does not prove It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success. [C003]

EVIDENCE BOUNDARY

A matching result does not prove It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

02

Threat Mechanics

Technical mechanism, observable transitions and supported scope.

ATTACK FLOW

TECHNICAL BACKGROUND

The retained dossier links Hollowgraph to the detection opportunity described as hollowgraph calendar c2. The v2 migration preserves the query exactly and exposes its match boundary, but does not reinterpret legacy source claims as laboratory evidence. Evaluate MicrosoftGraphActivityLogs for hollowgraph calendar c2 using the existing source-derived conditions while preserving projected investigation context. [C001][C002]

1

THREAT CONTEXT

access associated with a covert bidirectional channel. Observable: Context retained from the legacy dossier and source ledger.

2

OBSERVABLE CANDIDATE

Evaluate MicrosoftGraphActivityLogs for hollowgraph calendar c2 using the existing source-derived conditions while preserving projected investigation context. Observable: Events from MicrosoftGraphActivityLogs and projected fields.

3

CONTEXTUAL ENRICHMENT

Resolve asset, identity, process, ownership and change context before disposition. Observable: Raw event, surrounding timeline, inventory and authorized-change evidence.

4

DECISION GATE

Escalate only when corroborating evidence closes the documented non-claim boundary. Observable: Controlled validation, explainable negatives and analyst-readable evidence.

03

Telemetry & Evidence Model

Minimum data contract and known visibility gaps.

DATA CONTRACT

PRIMARY PLATFORM

Microsoft Defender XDR. Evaluate MicrosoftGraphActivityLogs for hollowgraph calendar c2 using the existing source-derived conditions while preserving projected investigation context.

FIELD / TABLE	STATUS	PURPOSE
MicrosoftGraphActivityLogs	**Required**	Primary event source selected by the unchanged query logic.
Event time and entity identifiers	**Required**	Chronology, deduplication, scoping and investigation pivots.
Raw projected context	**Required**	Preserve the fields emitted by the query before normalization or tuning.
Asset, identity and change inventory	**Enrichment**	Separate authorized activity and define operational ownership.

KNOWN VISIBILITY GAPS

Unvalidated local schema Missing projected fields Incomplete retention or asset scope Mutable attacker behavior Legacy claim provenance review pending

04

Primary Detection Logic

Native query, explicit condition evidence and match meaning.

KQL

```
// Rule: [T1102.002][T1071.001] Hollowgraph - Hollowgraph Calendar C2
// MITRE ATT&CK: T1102.002, T1071.001
// Classification: Production candidate
// Status: Production candidate
// Primary platform: Microsoft Defender XDR
// Required telemetry: MicrosoftGraphActivityLogs
// Purpose: access associated with a covert bidirectional channel.
// Logic: Evaluate MicrosoftGraphActivityLogs for hollowgraph calendar c2 using the existing source-derived conditions while preserving projected investigation conte
// A match proves: A match indicates successful Graph access to the exact far-future calendar range used by the analyzed sample.
// A match does not prove: It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.
// Expected false positives: authorized administration, approved software, testing, or benign artifact collisions.
// Coverage limits: local schema, field population, retention, collection scope, and attacker variation require validation.
// Evidence claims: C001, C002, C003
//
// Title: HOLLOWGRAPH calendar C2 access via Microsoft Graph
// Status: Production candidate
// Confidence: High
// Purpose: Detect successful Microsoft Graph calendarView requests to the
//          exact far-future date observed in the HOLLOWGRAPH sample.
// Required table: MicrosoftGraphActivityLogs
// Required fields: TimeGenerated, RequestMethod, RequestUri,
//                  ResponseStatusCode, AppId, ServicePrincipalId, UserId,
//                  ClientAuthMethod, IPAddress, UserAgent
// MITRE ATT&CK: T1102.002, T1071.001
//
// Important:
// - This is a production candidate, not a production-validated rule.
// - RequestUri must retain the startDateTime query parameter.
// - The rule deliberately does not inspect event subjects, attachment names,
//   or request-body content because MicrosoftGraphActivityLogs does not
//   document those values as available fields.
// - Validate legitimate long-range calendar applications before deployment.

MicrosoftGraphActivityLogs
| where TimeGenerated >= ago(1h)
| where RequestMethod == "GET"
| where RequestUri contains "/calendarView"
| where RequestUri contains "2050-05-13"
| where ResponseStatusCode between (200 .. 299)
| project
    TimeGenerated,
    AppId,
    ServicePrincipalId,
    UserId,
    ClientAuthMethod,
    IPAddress,
    UserAgent,
    RequestMethod,
    RequestUri,
    ResponseStatusCode
```

WHAT A MATCH PROVES

A match indicates successful Graph access to the exact far-future calendar range used by the analyzed sample.

WHAT IT DOES NOT PROVE

It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

MATURITY

Production candidate - Production candidate. Evaluate MicrosoftGraphActivityLogs for hollowgraph calendar c2 using the existing source-derived conditions while preserving projected investigation context.

05

Triage & Hunting Playbook

Evidence-preserving workflow from match to disposition.

WORKFLOW

01

Preserve the complete raw matching event, query version, timestamp, connector or sensor version and stable entity identifiers.

02

Confirm the event originated from the intended Microsoft Defender XDR data source and verify the semantics of every condition used by the query.

03

Resolve asset, identity, process, software and destination ownership against authoritative inventories and approved change records.

04

Build the surrounding endpoint, identity, network or appliance timeline and seek an independent evidence layer before escalation.

05

Classify benign overlap explicitly; tune only with durable ownership or provenance controls, never by suppressing the core behavior globally.

06

Escalate only when corroborating evidence establishes malicious or unauthorized activity beyond the documented match boundary.

DISPOSITION RULE

Escalate only when the evidence establishes the behavior described by the query. It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

06

False Positives & Validation

Overlap model, controlled checks and promotion gate.

QUALITY GATE

AUTHORIZED ADMINISTRATION OR SOFTWARE

Require owner, purpose, approved change and expected execution or configuration provenance.

BENIGN ARTIFACT OR COMMAND COLLISION

Require the full condition set plus an independent evidence layer or stable provenance.

PARSER, INVENTORY OR RETENTION AMBIGUITY

Reproduce a controlled event and reconcile raw fields, collection scope and authoritative inventory.

VALIDATION SEQUENCE

- Generate an authorized controlled equivalent of the primary observable without using live malicious infrastructure.
- Verify each query condition and projected field against retained raw telemetry.
- Run representative approved and near-match negative controls.
- Measure historical prevalence and classify every returned owner or workflow.
- Record parser, connector, retention and asset-scope dependencies with the test evidence.

SUCCESS

The controlled positive produces a complete, explainable row and representative negatives are separable through durable context.

REJECTION

Reject promotion when field semantics, collection scope, benign ownership or the core behavior cannot be established reproducibly.

PROMOTION GATE

Require claim-level source review, a retained controlled positive, representative negatives, measured prevalence and an accountable operational owner.

07

Analyst Assessment & References

Coverage statement, open questions and source record.

FINAL

CONCLUSION

This package now uses the canonical v2 layout and evidence contract while preserving the original detection.kql conditions. It remains a production candidate at high confidence. The visual migration is complete; claim-level provenance and target-environment execution remain explicit promotion gates.

MITRE ATT&CK

T1102.002 / T1071.001

PRIMARY AND AUTHORITATIVE REFERENCES

S001 group-ib.com - Group-IB - HOLLOWGRAPH: Turning Microsoft Graph into a Covert C2 Channel (undated)

hxxps://www[.]group-ib[.]com/blog/hollowgraph-microsoft-365/

S002 Microsoft Learn - Microsoft Learn - Microsoft Graph activity logs overview (undated)

hxxps://learn[.]microsoft[.]com/en-us/graph/microsoft-graph-activity-logs-overview

S003 Microsoft Learn - Microsoft Learn - Microsoft Graph activity logs overview (undated)

hxxps://learn[.]microsoft[.]com/en-us/azure/azure-monitor/reference/tables/microsoftgraphactivitylogs

S004 MITRE ATT&CK - T1102.002 - Web Service: Bidirectional Communication (undated)

hxxps://attack[.]mitre[.]org/techniques/T1102/002/

S005 MITRE ATT&CK - T1102.002 - Web Service: Bidirectional Communication (undated)

hxxps://attack[.]mitre[.]org/techniques/T1071/001/

OPEN QUESTIONS

A matching result does not prove It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

DOCUMENT CONTROL

Schema 2.0 | Layout 2.0 | Maturity production-candidate | Confidence High | No customer data | Evidence ledger required